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b) The forward reaction is endothermic

c) The reaction does not attain equilibrium

d) The reverse reaction is exothermic

4. The formation of ammonia from $N_2(g)$ and $H_2(g)$ is a reversible reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + \text{Heat}$$

What is the effect of increase of temperature on this equilibrium reaction

a) equilibrium is unaltered

b) formation of ammonia is favoured

c) equilibrium is shifted to the left

d) reaction rate does not change

5. Solubility of carbon dioxide gas in cold water can be increased by

a) increase in pressure

b) decrease in pressure

c) increase in volume

d) none of these

6 Which one of the following is incorrect statement ?

a) for a system at equilibrium, Q is always less than the equilibrium constant

b) equilibrium can be attained from either side of the reaction

c) presence of catalyst affects both the forward reaction and reverse reaction to the same extent

d) Equilibrium constant varied with temperature

7. K_1 and K_2 are the equilibrium constants for the reactions respectively.

$$N_2(g) + O_2(g) \overset{K_1}{\rightleftharpoons} 2NO(g)$$

$$2NO(g) + O_2(g) \overset{K_2}{\rightleftharpoons} 2NO_2(g)$$

What is the equilibrium constant for the reaction $NO_2(g) \rightleftharpoons \frac{1}{2}N_2(g) + O_2(g)$

a) $\frac{1}{\sqrt{K_1K_2}}$ b) $(K_1 = K_2)^{1/2}$

c) $\frac{1}{2K_1K_2}$ d) $\left(\frac{1}{K_1K_2}\right)^{3/2}$

8. In the equilibrium,

$$2A(g) \rightleftharpoons 2B(g) + C_2(g)$$

the equilibrium concentrations of A, B and C_2 at 400 K are 1×10^{-4} M, 2.0×10^{-3} M, 1.5×10^{-4} M respectively. The value of K_C for the equilibrium at 400 K is

a) 0.06 b) 0.09

c) 0.62 d) 3×10^{-2}

9. An equilibrium constant of 3.2×10^{-6} for a reaction means, the equilibrium is

a) largely towards forward direction

b) largely towards reverse direction

c) never established

d) none of these

10. $\frac{K_C}{K_P}$ for the reaction,

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is

a) $\frac{1}{RT}$ b) $\sqrt{RT}$

c) RT d) $(RT)^2$

11. For the reaction AB (g) $\rightleftharpoons$ A(g) + B(g), at equilibrium, AB is 20% dissociated at a total pressure of P, The equilibrium